THE UNITED REPUBLIC OF TANZANIA

THE PRESIDENT'S OFFICE

REGONAL ADMINISTRATION AND LOCAL GOVERNMENT

SHINYANGA REGION

031/2B **FORM FOUR MOCK EXAMINATION -2024**

PHYSICS -2B

Time: 2:30HRS **YEAR-2024**

INSTRUCTIONS.

1. This paper consists of **two (2)** questions. Answer all the questions.

2. Each questions carries twenty five **25 (marks)**

3. All writings should be in blue or black ink except for diagrams which must be in pencil.

4. Mathematical tables and non –programmable calculators may be used.

5. Communications devices and any authorized materials are not allowed in the examination room.

6. Write your Examination number on every page of your answer sheet **(s)** provided.

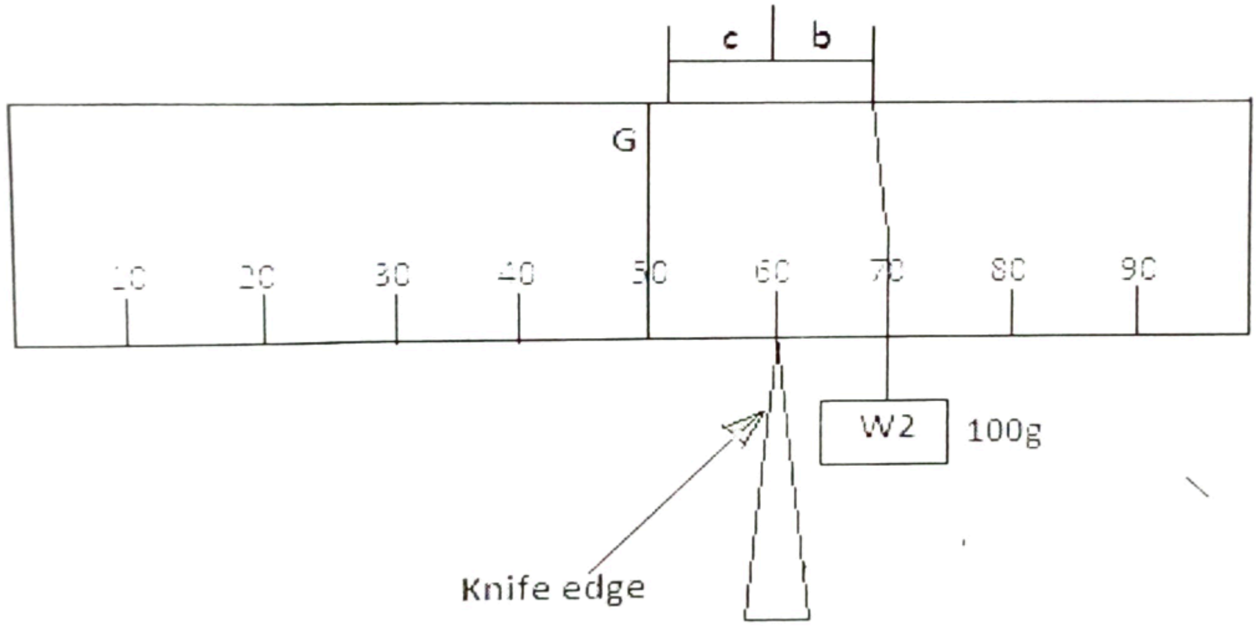
The following information may be useful

Pie =**3.14**

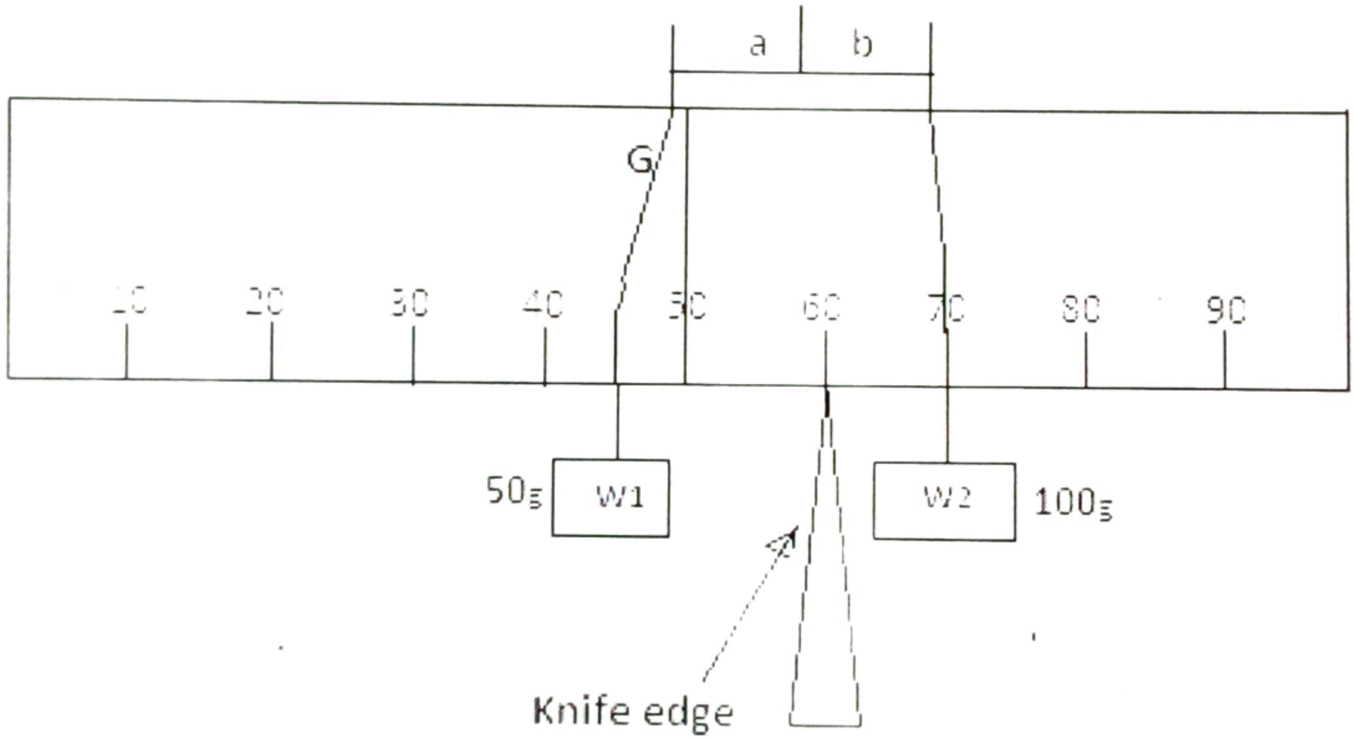
1.You are provided with meter rule ,a knife edge, and two strings of length 1**00cm** each and weight $\mathbf{W_1}$ and $\mathbf{W_2}$ of masses **50g** and **100g** respectively .Proceed as follows

(a) Balance a meter rule on a knife edge, put a mark and write **G** and the balancing point using a piece of chalk or a pencil .Measure and record the length **l,** Width **w** and the thickness **t** of a meter rule using a venier caliper.

(b) Place the meter rule on a knife edge so that the knife edge is at **60cm** of your meter rule see the figure 1 (a) .Suspend weight $\mathbf{W_2}$ **of 100g** on the right side of the knife edge. Adjust $\mathbf{W_2}$until the meter rule balances horizontally. Read and record lengths b and c as seen in figure 1 (a)



(i) Suspend weight W_1 **of 50g** on the left hand side of the knife edge at the position **47cm** and adjust weight W_2 until the meter rule balances horizontally as seen in figure 1 (b) . Read and record the lengths **a and b.**



(ii) Repeat the procedures in (b) (i) by adjusting the position of W_1 to the left at the interval of **3cm** to obtain other four (4) readings

(c) Tabulate your results as show in table below

a(cm)	**b(cm)**

(d) Plot the graph of **b** against **a**

(e) What is the nature of the graph?

(f) Calculate the slope **S** of the graph

(g) (i) Read the b-intercept **,** given that **b= Sa+(W/W_2)xc**

(ii) What does **(W_1/W_2) XC** represents in your graph?

(h) (i) calculate the value of W using the relation **W_2= (lxwxt)/m**

Note the mass of mate rule can be found by calculation

(ii) What is the physical meaning of the value **P?**

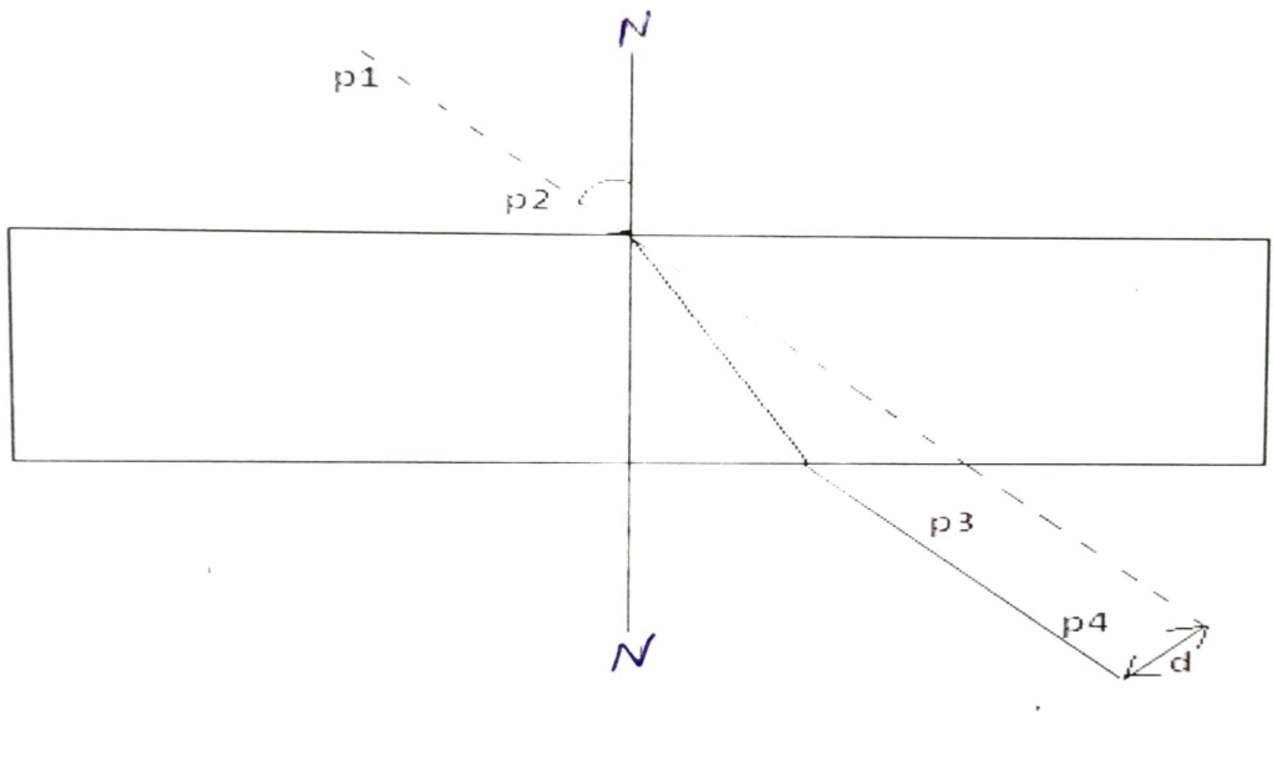
(iii) State the aim of this experiment.

(iv) State a possible source of error in this experiment.

(v) How can you minimize error in (**IV**) above?

2 You are provided with a glass block, four optical pins for sheets of drawing paper, and drawing board. Proceed as follows.

Place the glass block flat on the drawing paper fixed to the drawing board and with a sharp pencil draws its outline.



Remove the glass block and draw a normal NN' to the longer edge of the block, see fig. above

Draw a line making an angle of incidence **(i)** of 30^0.Stick two vertical pins $\mathbf{P_1}$ and $\mathbf{P_2}$ on this line. Replace the glass block. Stick two pins $\mathbf{P_3}$ and $\mathbf{P_4}$ on the other side of the block so that they appear to be in the same straight line with the images of $\mathbf{P_1}$ and $\mathbf{P_2}$ as through the block. Remove the block and draw the complete path of the ray entering and leaving the block. Measure the angle of the refraction **(r).**

Produce the incident ray as shown in figure above and measure the perpendicular distance d between the incident ray and the emergent ray.

Repeat this procedure for angles of incidence of $\mathbf{40^0}$**,** $\mathbf{50^0}$**,** $\mathbf{60^0}$ **and** $\mathbf{70^0}$. In each case draw the block again on a fresh part of the paper.

(a) Record your results in a table as follows

i(degree)	**r(degree)**	**d(cm)**	**dcosr(cm)**	**Sin(i-r)**
$\mathbf{30^0}$				
$\mathbf{40^0}$				
$\mathbf{50^0}$				
$\mathbf{60^0}$				
$\mathbf{70^0}$				

(b) Plot a graph of **dcos**r against Sin **(i-r)**

(c) Find the gradient of the graph

(d) Measure the width of the glass block.

(e) How is the gradient related with of the glass block?

NB. Hand in your diagrams together with your answer

The glass block must of dimensions Of (100x60x18)

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PHYSICS 2B MARKING SCHEME

Qn. 1.

(a) Table of results:

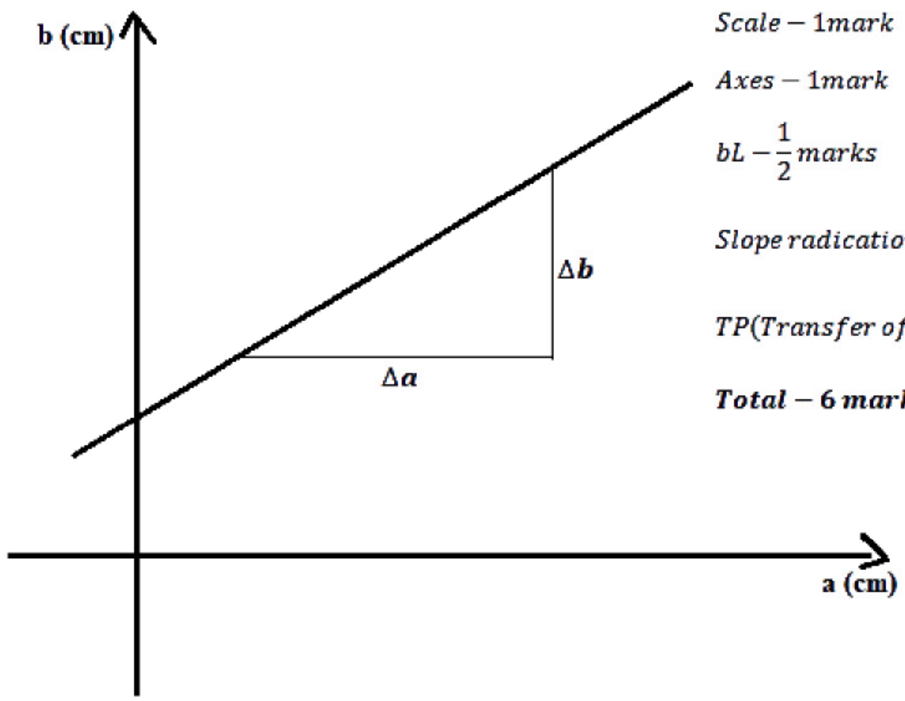
Length a (cm)	13	19	22	25	28
Length b (cm)	19.5	20.5	22.2	25	26.5

(5 Marks)

Length (L) = 100mm
Width (w) = 60mm
Thickness (t) = 18mm $\mathbf{1\frac{1}{2}}$ $(\boldsymbol{marks})$

(b) c = 10cm, b = 13 cm ***(1 Marks)***

Nature of the graph:



$Title - \frac{1}{2} marks$

$Scale - 1 mark$

$Axes - 1 mark$

$bL - \frac{1}{2} marks$

$Slope\ radication - \frac{1}{2} marks$

$TP(Transfer\ of\ point) - 2\frac{1}{2} marks$

$\boldsymbol{Total - 6\ marks}$

Nature of the graph is straight line through the b-intercept.

(c)

$$slope = \frac{\Delta b}{\Delta a} \;\; (^1/_2\, \boldsymbol{marks})$$

$$slope = \; 0.56 \; (^1/_2\, \boldsymbol{marks})$$

(d) (i) $b - intercept = 12.5\ cm \quad (^1/_2\, \boldsymbol{Mark})$

(ii) $\left(\frac{w}{w_c}\right) c\ \ Represents\ the\ b - intercept\ which\ is\ equal\ to\ 12.5\ cm \quad (^1/_2\, \boldsymbol{Mark})$

(iii) $Given\ the\ relation\ w_2 = \frac{w_c}{9.5cm}$

From the diagram no. 2

Sum of clockwise moment = sum of anticlockwise moment

$$w{:}\,10 = w_2\,.\,9.5\ cm \quad (\boldsymbol{2\ marks})$$

$$w = \frac{w_2.\,9.5\ cm}{100cm}$$

$$but\ w_2 = 100g$$

$$Then, w = \frac{100g \times 9.5cm}{100cm}$$

$$w = \; 95g \quad (\boldsymbol{1\ mark})$$

$$\therefore Mass\ of\ the\ metre\ rule\ used\ in\ the\ experiment = 95g$$

Note: The masses of the metre rule must be different because the metrerule have different masses

(e) (i) P=?

Given:

$$P = \frac{L \times W \times t}{M}$$

$$P = \frac{L \times W \times t}{M} \rightarrow \frac{100cm \times 2.5cm \times 0.5cm}{95g} (\boldsymbol{1\tfrac{1}{2}marks})$$

$$P = \frac{75cm^3}{95g} = \; 0.8\ cm^3/g\ (\boldsymbol{1mark})$$

(ii) The physical meaning of the value of P is the density of wood (metre rule used) ***(1 mark)***

(iv) Parallax error ***(2 marks)*** *one among them*

(v) Can be minimized by Reading the lengths perpendicular to the measurements of the metre rule ***(1 mark)*** *one among them*

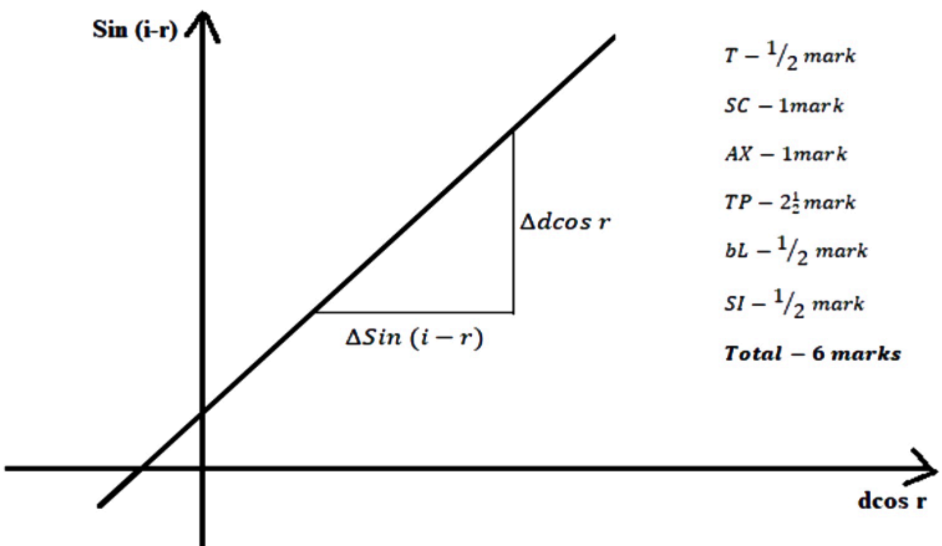
Qn.2

(a) Table of results (Table of values)

$i°$	$r°$	**d(cm)**	**d cos r**	**Sin (i-r)**
30	20	1.20	1.13	0.17
40	25	1.80	1.63	0.26
50	30	2.30	2.00	0.34
60	35	2.90	2.38	0.42
70	38	3.60	2.84	0.53

(12 Marks)

(b) Nature of the graph of dcos r against sin (i-r)



$T - {}^{1}/_{2}\ mark$

$SC - 1mark$

$AX - 1mark$

$TP - 2\frac{1}{2}mark$

$bL - {}^{1}/_{2}\ mark$

$SI - {}^{1}/_{2}\ mark$

$\boldsymbol{Total - 6\ marks}$

(c) From the graph

$$Slope = \frac{\Delta dcos\ r}{\Delta Sin\ (i-r)} \quad (\mathbf{1\ marks})$$

The slope is approximately to **6.4 cm*(1 marks)***

(d) The width of the glass block is 6.2 cm ***(2 marks)***

(e) The gradient of the graph is almost the same as the value of the width of the glass block

(3 marks)